

Discussion 3- Solutions

1- You are given the set of letters $\{A, B, C, D, E\}$. How many five-letter strings can be made if we require that each letter appears exactly once and the letters A and B are next to each other, as either “AB” or “BA”?

Problem 1 Solution.

There are three different way of solving this question. Learn one that feels most comfortable for you and use that one. My favorite one is method 2, I think it is simpler.

First method: They are asking how many permutations we can make with this letters and we want in those permutations A and B be always next together. For that I can think I have AB,C,D,E so I have four items and I want to calculate all the permutations with them and non is repetitive, so I have $4!$ permutations. But I could also have BA,C,D,E, (the order of A and B has changed), and I can have $4!$ permutations of this. So in general we have

$$4! + 4! = 2 \cdot 4! = 48$$

Second method: Or you could say I will count AB as one item, then in total I have 4 items, So $4!$ permutations, but Item AB is two guys in it which can be ordered in $2!$ ways, so in total I have

$$4!2! = 2 \cdot 4! = 48$$

Third method: Or you could say first I order C,D, E, so $3!$ ordering. Now think we have

$$_ C _ D _ E _$$

So there are 4 places that I can put my AB, so I have to choose 1 from this 4 places, $\binom{4}{1}$. So far we got $3! \cdot \binom{4}{1}$. But remember we can arrange A and B in $2!$ ways. so in total we have

$$3! \cdot \binom{4}{1} \cdot 2! = 3! \cdot 4 \cdot 2 = 4! \cdot 2 = 24 \cdot 2 = 48$$

2- Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is

a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

Problem 2 Solution.

Note that the solution is written for a more general case, which also includes the answer for your question. The purpose is for you to learn what to do in a more general case.

Your question was

2- Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book.

Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible? The answer is only $|E|$ look at the rest of solution.

I have written below the answer for updated 2

2-new - Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book.

What is the probability that Ms. Jones arranges her books so that all the books dealing with the same subject are together on the shelf. The answer is $P(E) = \frac{|E|}{|S|}$ see below.

Sample space = $S = \{\text{All the possible orderings of the book, with no restriction}\}$. So in total there are 10 books, so $|S| = 10!$.

$E = \{\text{Order the book in the way Books, same subjects are together}\}$. In total we have 4 subjects, so $4!$ ordering for subjects. But math books can be arranged in $4!$ way, chemistry book in $3!$, history books in $2!$ and language only in $1!$. so in total

$$|E| = \underbrace{4!}_{\text{subjects}} \cdot \underbrace{4!}_{\text{math}} \cdot \underbrace{3!}_{\text{Chemistry}} \cdot \underbrace{2!}_{\text{history}} \cdot \underbrace{1!}_{\text{language}}$$

Therefore,

$$P(E) = \frac{4!4!3!2!}{10!}$$

3- From a group of 5 women and 7 men, we want to choose a committee of 5 people.

- (a) Probability of choosing 3 women and 2 men?
- (b) Probability of choosing 3 women and 2 men if 2 of the men are feuding and refuse to serve on the committee together?

Problem 3 Solution.

- (a) $S = \{\text{choose 5 people from 12 people}\}$.
- $E = \{\text{choose 3 women from the 5 women and 2 men from the 7 men}\}$

$$P(E) = P(3 \text{ women and 2 men}) = \frac{\binom{5}{3} \binom{7}{2}}{\binom{12}{5}}$$

- (b) $E =$ First we will count number of ways that we can generally choose 3 women from 5 women and 2 men from 7 men. This is same as event E in part a, i.e.

$$E = \{\text{choose 3 women from the 5 women and 2 men from the 7 men}\}$$

Then now we have the scenario that we do not want these two men for example Jack and Alex to be chosen together, but each of them can be chosen with any other people. But let's assume we are choosing three women and these two men that are Jack and Alex together. Let's call it

$$B = \{\text{We are choosing three women from 5 women and 2 men from } \{\text{Alex, Jack}\} \}.$$

Now what you need is $|E| - |B|$. **Note that this is only complement of B with respect to E not to the whole sample space. Hence we get the correct answer by**

$$\frac{\binom{5}{3} \binom{7}{2} - \binom{5}{3} \binom{2}{2}}{\binom{12}{5}}$$

- 4. a. In how many ways can 3 boys and 3 girls sit in a row?
- b. In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?
- c. In how many ways if only the boys must sit together?
- d. In how many ways if no two people of the same sex are allowed to sit together?

- a. no restriction so $6!$
- b. girls among themselves $3!$, boys among themselves $3!$, and $2!$ for girl boy or boy girl.

Hence, in total $3! \cdot 3! \cdot 2!$

- c. We can assume all boys together form 1 squeezed group Boys, hence 1 Boys(whole squeezed group) + 3 girls, in total we have 4 people hence $4!$, but boys among themselves can be rearranged in $3!$, hence in total we have $4! \cdot 3!$
- d. we could only have patterns like

$$b_1 g_1 b_2 g_2 b_3 g_3$$

or

$$g_1 b_1 g_2 b_2 g_3 b_3.$$

- $2!$ for whether we start with a boy or girl.
- $3!$ for boys
- $3!$ girls

in total $2! \cdot 3! \cdot 3!$

5. If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? For this question assume a year has 365 days.

The sample space size: If there are no restriction for each person any of the 365 days could be its birthday. So for each person we have 365 days and in total we have n people so it is 365^n .

For the event when no two people should have the same birthday, we could say for the first person we have 365 days that could be their birthday. Once that is selected, for the second person they should not have the birthday same as the first one, so only 364 options are available, and we continue the same way hence,

$$\frac{365 * 364 * \cdots * (365 - (n - 1))}{365^n}$$

6. Suppose $P(B) = 0.45$, $P(A \cup B) = 0.67$, $P(A \cup B') = 0.89$. Calculate the followings

- (a) $P(A)$
- (b) $P(A|B)$
- (c) $P(A|B')$

Problem 4 Solution.

$$P(A \cup B) - P(B) = P(A \cap B') = 0.67 - 0.45 = 0.22$$

$$P(A \cup B') - P(B') = P(A \cap B) = 0.89 - 0.55 = 0.34$$

$$P(A) = P(A \cap B') + P(A \cap B) = 0.22 + 0.34 = 0.56$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.34}{0.45} = \frac{34}{45}$$

$$P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{0.22}{0.55} = \frac{22}{55}$$